

Week 1 Lab Assistant Preparation

GitHub Classroom, Git basics, computer hardware, and Raspberry Pi hardware identification.

Main focus

For Week 1, your main job is helping students get started with Git/GitHub and understand basic hardware. Raspberry Pi appears, but mainly for identifying physical parts, not deep terminal work yet.

Quick overview

Area	What students do	What you should be ready for
Git/GitHub	Accept assignment, clone repo, edit Markdown, commit and push.	Fix login issues, wrong folders, missing push, and explain Git workflow simply.
Hardware vs software	Write short explanations of hardware, software, and interaction.	Give beginner examples and explain OS, files, drivers, and programs.
Computer disassembly	Inspect a computer, take photos, identify internal parts.	Safety checks, component identification, photo/image path support.
Raspberry Pi identification	Identify SoC, GPIO, USB, HDMI, power, microSD, network, wireless.	Explain Pi parts simply and point them out on the board.
Reflection	Write 200-300 words about what they learned.	Help them write specific reflections, not generic statements.

Relevant repo files

```
week01/01_git/01_git_github.md
week01/02_computers/01_hardware_vs_software.md
week01/02_computers/02_computer_disassembly.md
week01/02_computers/03_component_investigation.md
week01/02_computers/04_raspberry_pi_identification.md
week01/02_computers/05_reflection.md
```

1. GitHub Classroom and Git

Students need to accept the GitHub Classroom assignment, clone their generated repository, edit files in VS Code, then commit and push their changes back to GitHub.

Commands you should know well

```
git clone <repo-url>
cd <repo-folder>
git status
git add .
git commit -m "Completed week 1"
git push
```

Beginner explanation

clone = download repo first time
 status = check what changed
 add = choose changes to save
 commit = save a checkpoint locally
 push = upload checkpoint to GitHub
 pull = download latest changes

Common Git mistakes and fixes

Problem	Likely cause	How to fix
fatal: not a git repository	They are running Git commands outside the repo folder.	Run pwd, then ls, then cd into the correct repo folder.
Nothing appears on GitHub	They saved locally but did not push.	Run git status, git add ., git commit, then git push. Check GitHub website.
nothing to commit, working tree clean	No saved changes, wrong folder, or already committed.	Check file is saved. Run pwd and ls. Confirm they are inside the repo.
Authentication failed	GitHub login/token/credential issue.	Use browser login if prompted. Confirm correct GitHub account and accepted Classroom repo.
Edited on GitHub website and locally	Local copy may be behind remote version.	Use git pull before continuing. Help carefully if a conflict appears.

Important sentence to repeat

Saving in VS Code is not submission. Commit saves locally. Push uploads to GitHub. If it is not visible on GitHub.com, it is probably not submitted.

2. Hardware vs Software

Students must explain what hardware is, what software is, and how they interact. Keep explanations concrete and beginner-friendly.

Concept	Simple explanation	Examples
Hardware	Physical parts of a computer that you can touch.	CPU, RAM, keyboard, monitor, motherboard, storage.
Software	Programs and instructions that tell hardware what to do.	Raspberry Pi OS, Windows, browser, VS Code, Python.
Interaction	Software sends instructions through the operating system to control hardware.	Typing on a keyboard appears on screen through OS and app interaction.

Fast answers for common questions

Student question	Good short answer
Is the operating system hardware or software?	Software. It manages hardware and lets programs run.
Is a file hardware or software?	It is data stored on hardware and managed by software.
Is Raspberry Pi hardware?	Yes. It is a small physical computer board.

Is Python hardware or software?	Software. It is a programming language and runtime.
What is a driver?	Software that helps the operating system talk to hardware.

3. Computer Disassembly

Students inspect a computer before opening it, take a photo, open the case safely, identify components, then take another photo.

Expected image paths:
 images/system/computer-before.jpg
 images/system/computer-opened.jpg

Safety points to repeat

- Turn off the computer before opening it.
- Unplug the power cable.
- Do not open the power supply unit.
- Do not force parts out.
- Avoid touching exposed circuits unnecessarily.
- Ask staff before removing any component.

Most important safety warning

Do not open the PSU. Power supplies can be dangerous even when the computer is unplugged.

What students can describe

Before opening	Inside the computer
desktop/laptop form factor, brand/model, visible ports, case size, front/back connectors	motherboard, CPU cooler, RAM sticks, storage drive, power supply, cables, fans, expansion cards

4. Component Investigation

Students fill a table for CPU, RAM, Storage, Motherboard, Power Supply, and Cooling. You should be ready to explain each in one simple sentence.

Component	Function	Specification examples
CPU	Processes instructions. Often called the brain of the computer.	Intel i5, AMD Ryzen, clock speed, number of cores.
RAM	Temporary working memory used while programs are running.	8GB DDR4, 16GB DDR5.
Storage	Long-term place where OS, apps, and files stay.	256GB SSD, 1TB HDD.
Motherboard	Main circuit board that connects all components.	Model, chipset, ports, expansion slots.
Power Supply	Converts wall power into power the computer can use.	300W, 500W, 80 Plus rating.
Cooling	Removes heat from components.	Fans, heatsink, thermal paste.

RAM vs storage analogy

RAM is like your desk while working. Storage is like your cupboard where things stay after shutdown.

5. Raspberry Pi Hardware Identification

In Week 1, Raspberry Pi work is mainly physical identification. Students should recognise the main ports and parts on the board.

Pi part	Beginner explanation
SoC	Main chip containing CPU and other important controllers.
GPIO	Pins used to connect LEDs, buttons, sensors, motors, and other hardware.
USB	Ports for keyboard, mouse, storage, and accessories.
HDMI	Connects the Pi to a monitor or display.
Power	Where the Pi receives power, often USB-C on newer models.
microSD	Storage card containing the operating system and files.
Network	Ethernet port for wired internet/network connection.
Wireless	Built-in Wi-Fi and Bluetooth capability.

Common Raspberry Pi confusion

Confusion	Simple explanation
SoC vs CPU	A CPU is the processor. An SoC is a bigger chip that includes CPU plus other components.
Why microSD matters	On Raspberry Pi, the microSD card acts like the main storage drive and the OS boots from it.
What GPIO is for	GPIO lets the Pi interact with real-world devices like LEDs, sensors, buttons, and motors.

6. Reflection Support

Students need to write a 200-300 word reflection. Help them avoid generic statements and write specific observations.

Weak reflection	Better structure
I learned about computers. It was interesting.	Before this lab, I thought... During the activity, I noticed... The most difficult part was... I now understand that... In embedded systems like Raspberry Pi...

7. Physical and Software Setup Checklist

Check	Why it matters
Git installed on lab machines	Students need terminal Git commands.
VS Code installed	Students need to edit Markdown files.

GitHub Classroom link available	Students need assignment access.
Students have GitHub accounts	Delays happen if they create accounts during lab.
Old computers available	Needed for disassembly task if done physically.
Screwdrivers available	Needed to open cases safely.
Raspberry Pis available	Needed for board identification.
Photo policy clear	Students may need to take and upload photos.
Image folders clear	Students need to know where screenshots/photos go.

8. Your Before-Lab Checklist

- Clone a repo and open it in VS Code.
- Edit a Markdown file, commit it, and push it.
- Explain clone, status, add, commit, push, and pull.
- Fix fatal: not a git repository.
- Explain hardware vs software with examples.
- Explain CPU, RAM, storage, motherboard, PSU, and cooling.
- Identify Raspberry Pi SoC, GPIO, USB, HDMI, power, microSD, Ethernet, and wireless.
- Show how to add an image in Markdown.
- Repeat the PSU safety warning.
- Ask staff what counts as completed Week 1 work.

9. Missing or Unclear Items to Clarify with Staff

- Are students expected to upload actual photos into the repo?
- Should students create the images folder themselves?
- Are students allowed to remove components or only observe them?
- Is the GitHub Classroom repo individual or team-based?
- What exactly counts as completed Week 1 work?
- Are students expected to use Raspberry Pi physically in Week 1, or only identify labelled parts?

Final reminder

For Week 1, focus on Git/GitHub setup and beginner hardware understanding. Most student issues will be Git workflow problems, wrong folders, image placement, and basic component confusion.